

Walmart Sales Forecasting

Using Facebook Prophet • Time Series Analysis Report

Dataset: Rossmann Store Sales | Stores: 1,115 | Period: Jan 2013 – Jul 2015

1. Objective

The objective of this project is to forecast future sales using historical Walmart store sales data, and to identify underlying trends and seasonal patterns. The model provides actionable insights that support inventory planning, resource allocation, and strategic business decision-making.

2. Dataset Overview

Attribute	Detail
Source	Walmart Store Sales Forecasting Dataset
Total Records	1,017,209 rows
Stores	1,115 unique stores
Date Range	January 2013 – July 2015
Target Variable	Daily Sales (aggregated across all open stores)
Key Features	Store, Date, Sales, Customers, Open, Promo, StateHoliday, SchoolHoliday

3. Methodology

- **Data Collection & Preprocessing** — Loaded raw CSV, parsed dates, filtered closed-store days (Open=0), and aggregated daily sales across all stores.
- **Exploratory Analysis** — Examined historical trends, weekly patterns, holiday effects, and sales distributions.
- **Train / Test Split** — 85% of data used for training (Jan 2013 – ~Mar 2015), 15% held out for evaluation.

- **Prophet Model Development** — Configured with yearly and weekly seasonality, `changepoint_prior_scale=0.1` to control trend flexibility.
- **Future Forecasting** — Generated a 90-day out-of-sample forecast beyond the last observed date.
- **Evaluation** — Assessed model accuracy using MAE, RMSE, and SMAPE on the held-out test set.
- **Visualisation** — Plotted historical sales, forecast vs actual, long-term trend component, and weekly seasonality patterns.

4. Tools & Technologies

Tool	Version/Role
Python	Core programming language
Pandas	Data loading, aggregation, preprocessing
Prophet	Time-series forecasting model (Meta/Facebook)
Matplotlib	Visualisation of forecasts and components
NumPy	Numerical operations and metric calculation
Scikit-learn	MAE and RMSE evaluation metrics

5. Model Performance

1,558,766	2,212,147	46.0%	23.5%
Mean Absolute Error	Root Mean Sq. Error	SMAPE	MAE / Mean Sales

The MAE of ~1.56 million reflects the aggregate scale of 1,115 stores. As a proportion of mean daily sales (~6.6 million), the error represents approximately **23.5%**, indicating reasonable predictive performance for aggregate-level retail forecasting. The elevated SMAPE (46%) is partially driven by low-sales Sundays when most stores are closed.

6. Forecast Visualisation



Figure 1: Historical sales, forecast vs actual, long-term trend, and weekly seasonality pattern.

7. Key Findings

Long-Term Growth Trend

Sales exhibited a steady upward trajectory from 2013 to mid-2015, with a noticeable acceleration in mid-2014 captured by Prophet's changepoint detection.

Weekly Seasonality

A strong weekly pattern was identified — Monday and Friday are peak sales days, while Sunday is significantly depressed due to store closures.

Seasonal Fluctuations

Yearly seasonality was successfully modelled, capturing holiday-period peaks and mid-year dips consistent across all three years of data.

90-Day Projection

The model's forward projection from August 2015 continues the established seasonal and trend patterns, providing a reliable near-term sales outlook.

Confidence Intervals

The 80% confidence band widens appropriately into the future, honestly representing increasing forecast uncertainty over time.

8. Recommendations for Improvement

- **Per-Store Models** — Training individual Prophet models per store would capture store-level idiosyncrasies and likely reduce SMAPE significantly.
- **External Regressors** — Adding Promo, SchoolHoliday, and StateHoliday flags as additional regressors in Prophet can improve accuracy on holiday-affected periods.
- **Log Transform** — Applying a log transform to sales before modelling reduces the influence of large-scale variance and often improves MAPE.
- **Cross-Validation** — Prophet's built-in time-series cross-validation (`prophet.diagnostics`) would provide more robust metric estimates than a single split.
- **Ensemble Approaches** — Combining Prophet with LightGBM or XGBoost (using lag features) typically yields superior results for retail forecasting.

9. Conclusion

The Facebook Prophet forecasting model demonstrated effective performance on aggregate Walmart store sales data. It successfully decomposed the time series into interpretable trend and seasonality components, and generated a credible 90-day forward forecast with quantified uncertainty. With an MAE-to-mean ratio of 23.5%, the model provides a solid baseline for retail sales forecasting. The insights derived — including weekly demand patterns and long-term growth trajectory — can directly inform inventory planning, staffing decisions, and promotional strategy.